

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 - SUB-STRAND 1.2: THE CELL - FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: What makes up an organism, and how do the tiny structures inside a single cell keep a whole living thing, from a sukuma wiki plant to a human being in Nairobi, alive? Remember the Kisumu learner who left a sukuma wiki leaf in the sun. It wilted and died. Under the school light microscope, the learner saw box-like compartments with green granules. Under the electron microscope, the same leaf cell showed folded membranes, dark oval bodies and a large fluid-filled space. Use what you observed and learned in Lessons 1 to 8 to answer each question below in your own words. Write your answers in the space under each prompt. Use correct science words and give reasons for your ideas. Where a diagram is asked for, draw and label it neatly. You may use your cell model, your notes and the Driving Question Board.

Part 1: Two microscopes, two pictures

The Kisumu learner was puzzled that the light microscope and the electron microscope showed such different pictures of the same sukuma wiki leaf cell. (a) Explain why the two microscopes show different levels of detail. Refer to resolution and magnification. (b) Name two other differences between a light microscope and an electron microscope. (c) Which structures were visible only under the electron microscope in the sukuma wiki cell?

Write your answer here:

Part 2: Plant and animal cells

(a) Draw and label a plant cell as seen under the electron microscope. Include at least eight labels. (b) State three structures found in the sukuma wiki cell but not in a human cheek cell. (c) State one structure found in the human cheek cell but not in the plant cell.

Write your answer here:

Part 3: What each structure does

Choose any five organelles from the sukuma wiki cell or the human cheek cell. For each one, state its function. Then use Kenyan examples, such as a jiko, ugali or githeri, to explain how at least two of these organelles depend on each other.

Write your answer here:

Part 4: Why the leaf wilts and dies

Use what you know about organelles to explain, step by step, why the sukuma wiki leaf left in the sun wilts and eventually dries and dies. Also explain why a fresh leaf is green and feels crisp.

Write your answer here:

Part 5: From organelle to organism

(a) Arrange these levels of organisation from the smallest to the largest: organ, organelle, organism, cell, organ system, tissue. (b) Give one Kenyan example for each of the sukuma wiki plant and a human being, showing all six levels. (c) Explain why no single organelle or cell can keep a whole organism alive. Then answer the driving question in two or three sentences.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining the two microscopes	Clearly links the different pictures to resolution and magnification, using the correct values or comparisons, and names two further differences correctly.	Says the electron microscope shows more detail and gives one or two correct differences, but without clear reference to resolution.	States that one microscope is better, with no correct scientific reason or differences.
Cell structure and diagram	Accurate, neat labelled plant cell with eight or more correct labels and correct plant versus animal differences.	Mostly accurate diagram with five to seven correct labels and at least two correct differences.	Diagram is incomplete or has several wrong labels and the differences are mostly incorrect.
Organelle functions and interdependence	Gives correct functions for five organelles and explains how at least two depend on each other, using Kenyan examples.	Gives correct functions for most organelles and mentions a link between them.	Gives few correct functions and does not show how organelles work together.
Explaining the phenomenon	Gives a full, correctly ordered cause-and-effect chain from water loss and turgor to photosynthesis, mitochondria and death of cells, tissues and the leaf, and explains the green colour and crispness.	Explains wilting using turgor or photosynthesis, with some steps missing.	Only says the leaf dies because it dries, with no reference to cell structures.
Levels of organisation and answering the driving question	Orders all six levels correctly, gives correct examples for both organisms, and answers the driving question using evidence.	Orders the levels with one or two errors and gives a partly complete answer.	Levels are mostly wrongly ordered and the answer to the driving question is missing or unrelated.

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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